REAL-TIME STOCK DASHBOARD & FORECASTING

PROJECT OVERVIEW

- **What it is :** An interactive web application built with Streamlit that provides users with tools for real-time stock data visualization and forecasting.
- **Purpose :** To empower users with the ability to monitor real-time stock prices, visualize historical trends, and make informed predictions about future stock movements.
- **Key Features :** Real-time stock prices

  Interactive charts (candlestick & line)

  Key metrics (last price, change, high/low, volume)

  Technical indicators (SMA, EMA)

  Stock forecasting with Prophet

TECHNOLOGIES USED

- **Streamlit:** A powerful open-source Python library for creating interactive web applications with minimal effort. It simplifies the process of building data-driven dashboards and apps.

- **Plotly:** An interactive graphing library that allows users to create visually appealing and dynamic charts. It enables zooming, panning, and hovering to explore data points in detail.

- **Pandas:** A versatile data manipulation and analysis library that provides data structures like DataFrames for efficient handling and analysis of stock data.

TECHNOLOGIES USED

- **yfinance:** A library that allows seamless retrieval of historical and real-time stock data from Yahoo Finance, providing a reliable data source for the application.

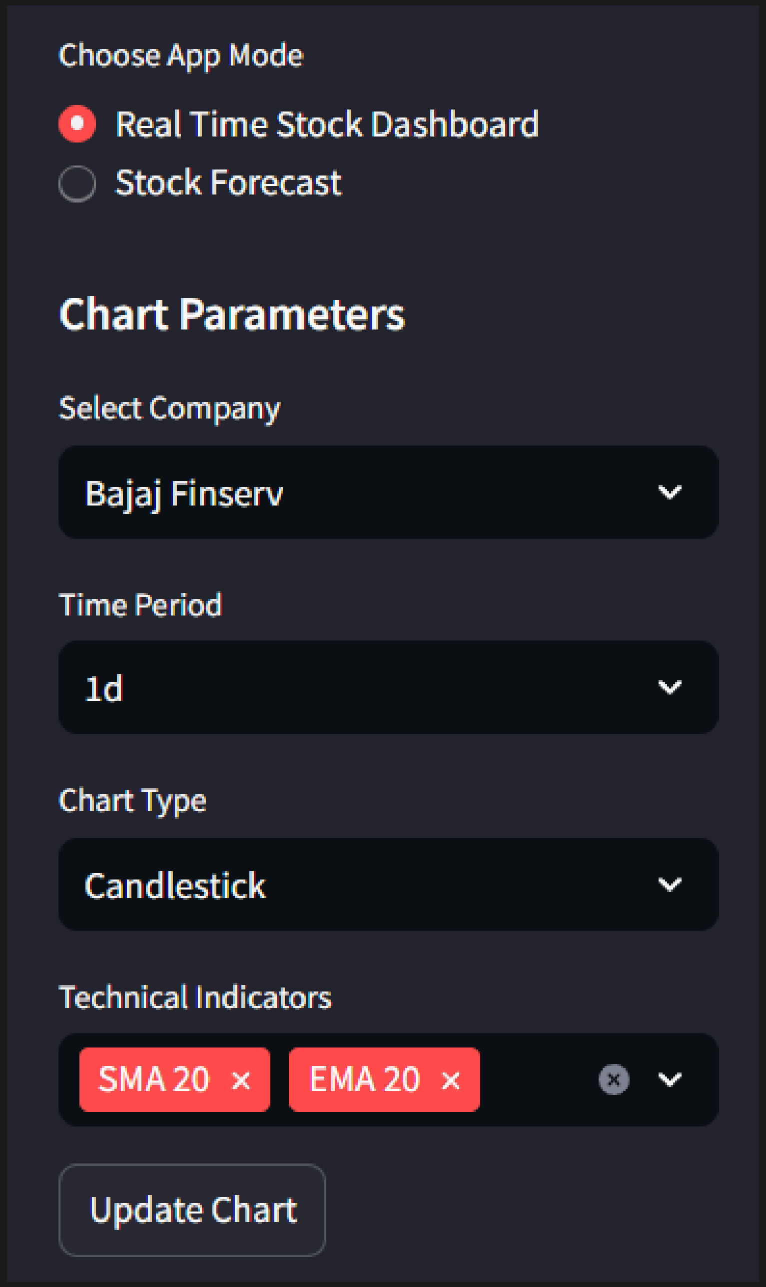
- **Prophet:** A robust time series forecasting model developed by Meta, known for its ability to effectively handle seasonality and trends in stock data.

- **TA-Lib:** A library that provides a wide range of functions for calculating various technical indicators, aiding in technical analysis of stock trends.

REAL-TIME STOCK DASHBOARD

Functionality : Select company from Nifty 50 list

- Choose time period (1d, 1wk, 1mo, 1y, max).
- Select chart type (candlestick, line).
- Add technical indicators (SMA 20, EMA 20).



DATA HANDLING

fetch_stock_data function:

- Utilizes the yfinance library to fetch historical stock data from Yahoo Finance based on the provided ticker, period, and interval.
- Employs the @st.cache_data decorator to cache the fetched data for 60 seconds, improving performance by reducing redundant data requests.

process_data function:

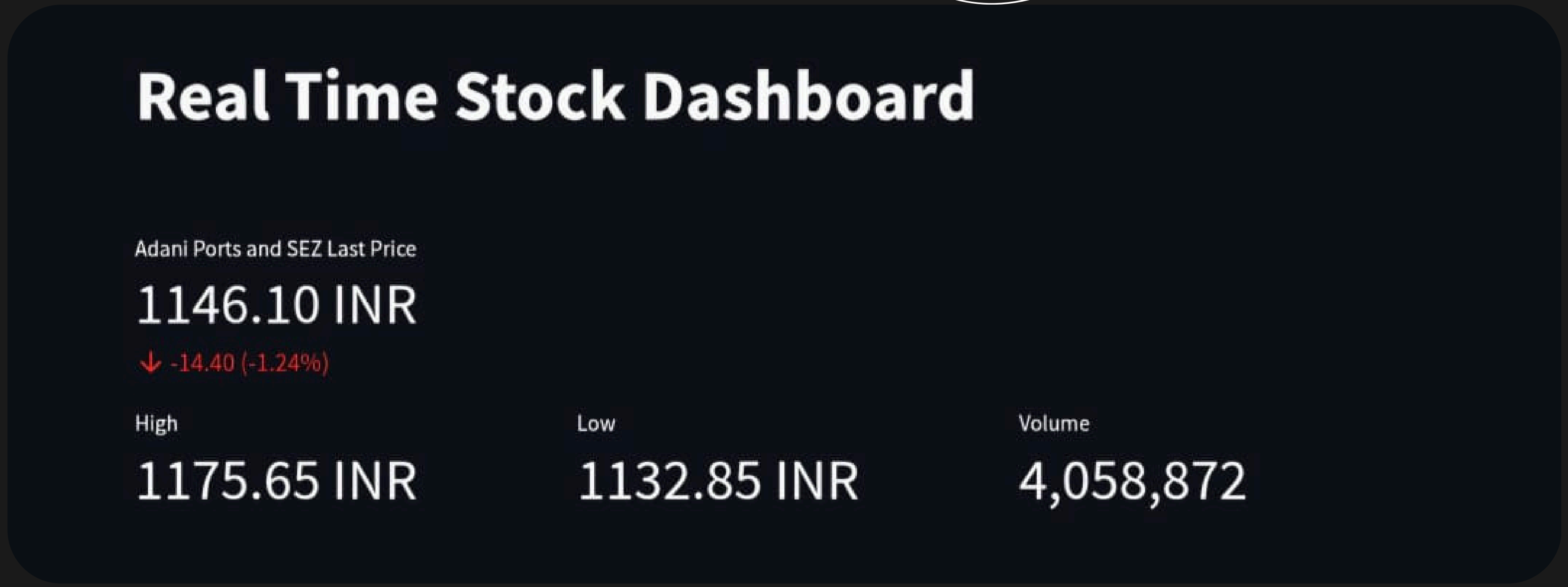
- Cleans and preprocesses the raw stock data to ensure consistency and accuracy.
- Handles MultiIndex columns, converts the datetime index to the Asia/Kolkata timezone, and removes rows with missing 'Close' values.

KEY METRICS

calculate_metrics function

Calculates and displays key metrics derived from the stock data:

- Last close price
- Price change and percentage change
- High and low prices
- Volume



TECHNICAL INDICATORS

add_technical_indicators function:

- Enhances the stock data by calculating and adding two commonly used technical indicators: 20-period Simple Moving Average (SMA) and 20-period Exponential Moving Average (EMA).

- Leverages the ta library for efficient calculation of these indicators.

STOCK FORECASTING WITH PROPHET

Prophet Model :

- A powerful time series forecasting model developed by Meta, designed to handle the complexities of stock market data.
- Excels at capturing seasonality and trend patterns, making it suitable for predicting future stock price movements.

fetch_stock_data_daily function:

- Specifically retrieves daily stock data using yfinance, as the Prophet model typically works best with daily granularity for forecasting.

FORECASTING PROCESS

forecast_pct_change_prophet function:

1. Prepares the data for Prophet by renaming columns to 'ds' (datestamp) and 'y' (value), as required by the model.
2. Fits the Prophet model to the historical stock data, enabling it to learn patterns and trends.
3. Makes future predictions for the specified number of forecast_days, providing insights into potential stock movements.
4. Returns the forecast results, including the predicted values ('yhat'), along with lower and upper bounds ('yhat_lower' and 'yhat_upper') representing uncertainty.

FORECAST VISUALIZATION

- Presents the forecasted percentage change values in a clear tabular format, allowing users to easily interpret the predictions.
- Utilizes Plotly to create an interactive chart that visualizes the forecast, with error bars representing the uncertainty intervals.

mse	rmse	mae	mdape	smape
1.1659	1.0798	0.8755	1.0272	1.7082
1.1758	1.0843	0.8644	1.0221	1.6876
1.285	1.1336	0.9181	1.0292	1.742
1.3144	1.1465	0.9301	1.0296	1.724
1.2625	1.1236	0.8961	1.0296	1.6994
1.2889	1.1353	0.892	1.0165	1.6431
1.5495	1.2448	0.937	1.0075	1.6339
1.4333	1.1972	0.8689	1.0075	1.5925
1.3921	1.1799	0.857	1.0153	1.5991

Perfomance Metrics

Choose App Mode

- Real Time Stock Dashboard
- Stock Forecast

Select Company

Bajaj Finserv

Number of Days to Forecast

3

Historical Data Period

1y

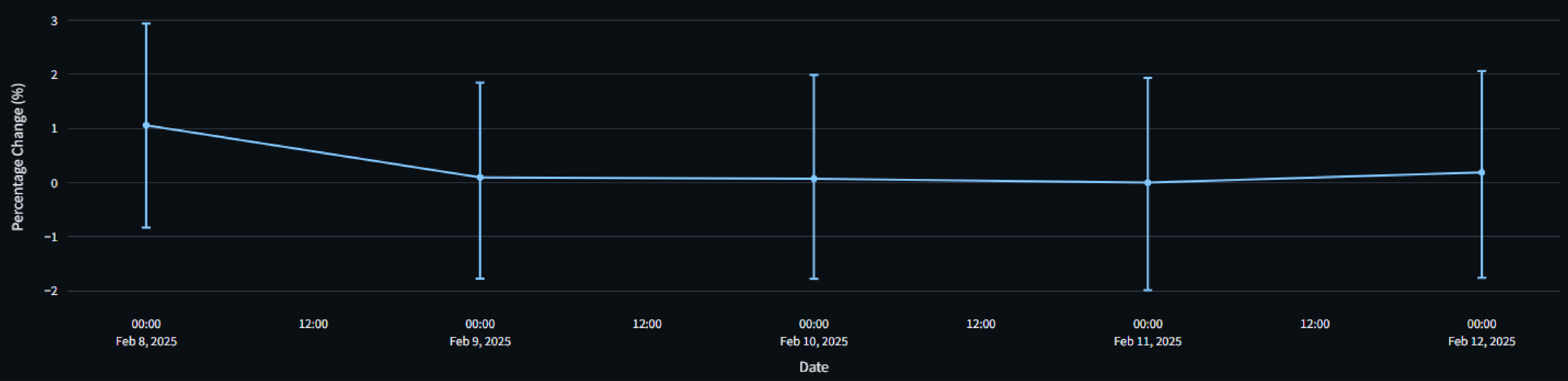
Run Forecast

Stock Forecasting

Forecasted Percentage Change for Next 3 Days

	Date	% Change	Lower Bound	Upper Bound
1	2025-02-08 00:00:00	1.0634	-0.8287	2.9428
2	2025-02-09 00:00:00	0.0992	-1.7719	1.8483
3	2025-02-10 00:00:00	0.0746	-1.7747	1.9909
4	2025-02-11 00:00:00	0.0011	-1.986	1.9384
5	2025-02-12 00:00:00	0.189	-1.7578	2.0627

Forecast of Bajaj Finserv Percentage Change



AI-POWERED INSIGHTS

- **AI Enhancement:** The app uses AI to provide deeper stock analysis beyond basic data and forecasts.
- **Gemini Integration:** The Gemini Vision model analyzes stock trends to give trading suggestions.
- **Visual Analysis:** Gemini interprets stock data visually, similar to a human analyst, using charts created by the get_image function.
- **Prompt-Driven Insights:** A detailed prompt guides Gemini to provide buy/hold/sell recommendations, summaries, and specific trading suggestions.
- **Seamless Integration:** AI-generated suggestions are displayed alongside other app features, enhancing the user experience.

STREAMLIT APP STRUCTURE

App Mode Selection :
Employs st.sidebar.radio to provide a user-friendly way to switch between the two main functionalities of the application:

- Real-time stock dashboard
- Stock forecasting

Conditional Logic :
Based on the user's mode selection, the application dynamically displays the relevant content and executes the corresponding code blocks.

USER INTERFACE

Sidebar Controls:

Offers intuitive controls in the sidebar for user interaction:

- Company selection dropdown to choose the desired stock.
- Time period selection for historical data visualization.
- Chart type selection (candlestick or line).
- Indicator selection to add SMA and EMA.
- Forecast horizon and period selection for customizing predictions.

Main Content Area:

- Dynamically displays the selected content, including charts, metrics, and forecast results, providing a clear and organized view of the information.

PROJECT STRENGTHS

- **Real-time data and interactivity:** Provides up-to-date stock information and allows users to interact with charts and data.
- **Comprehensive stock analysis tools:** Offers a variety of features for analyzing stock trends, including key metrics and technical indicators.
- **Forecasting capabilities with Prophet:** Integrates the powerful Prophet model to generate insightful stock price predictions.
- **User-friendly interface:** Presents a clean and intuitive interface, making it easy for users to navigate and utilize the application's functionalities.

FUTURE IMPROVEMENTS

- **Enhanced error handling:** Implement more robust error handling mechanisms to gracefully handle potential issues like network errors or invalid ticker symbols.
- **More technical indicators:** Expand the range of available technical indicators to provide users with additional tools for in-depth stock analysis.
- **Alternative data sources:** Explore integrating data from other sources besides Yahoo Finance to offer more comprehensive stock information.
- **Advanced forecasting models:** Experiment with and incorporate more advanced forecasting models, potentially including machine learning techniques, to improve prediction accuracy.
- **Deployment to a cloud platform:** Deploy the application to a cloud platform like Streamlit Cloud, Heroku, or Docker to make it accessible to a wider audience.

THANK YOU!